



DZPHARM • THE COMPLETE SYNTHESIS

The Final Execution Plan

The single source of truth. Every issue (122), every idea (200+), every design rule, every feature verdict, every verification check — from 10 source documents + 5 uploaded v2 files + 59 deep-dive PDFs + 6 technical designs — merged into one sequenced, classified, executable plan. Written for an AI agent to follow step by step. Nothing is missed. Everything is clear.

122

REGISTER ITEMS

200+

FEATURE IDEAS

14

VERIFY-FIRST CHECKS

8

EXECUTION PHASES

SOURCES SYNTHESIZED

5 uploaded v2 HTML files • 59 deep-dive PDFs (~600p)
6 technical designs (169p) • Blueprint + Sprint Plan + Ultimate
Blueprint
Design Universe + Name & Homepage Vision • Total: ~1,000+ pages

Instructions for the Executing AI Agent

This document is the **single source of truth** for transforming DzPharm from B- grade to A+. It synthesizes every source document — 5 uploaded v2 HTML files, 59 deep-dive PDFs, 6 technical designs, the transformation blueprint, sprint plan, ultimate blueprint, design universe, and name & homepage vision. **Read and execute in order.** Do not skip phases. Do not reorder. Every phase has dependencies on the previous one.

CRITICAL RULES FOR THE AI AGENT

- 1. VERIFY FIRST.** Run Phase 0 (14 checks) before touching any code. Reports contradict each other. Planning on wrong evidence wastes engineering time.
- 2. SAFETY CLUSTER FIRST.** No feature work starts ahead of the P0 safety cluster (P0-01 through P0-08). A second storey on an uncured foundation is how platforms die.
- 3. SEVERITY COLOURS ARE LOCKED.** The 7 clinical severity colours (safe/warning/danger/severe/info/none/unknown) stay locked outside every palette, theme, and animation. Decoration never shares a hue with a verdict. This is P0-07.
- 4. THE COPILOT DOES NOT DO ARITHMETIC.** An LLM computing doses with no deterministic cross-check is the single largest harm vector (P0-08). Every dose calculation must be verified by a deterministic engine, not the LLM.
- 5. ADDITIVE ONLY.** No rewrites. Every fix is additive on the existing Next.js + Tailwind + Zustand foundation. If a fix requires a rewrite, flag it and seek human guidance.
- 6. SHIP BEHIND FEATURE FLAGS.** Every change ships behind a feature flag. Roll out 10% → 50% → 100%. Monitor error rates. Roll back if >0.1%.
- 7. DOCUMENT EVERY CHANGE.** Append to /home/z/my-project/worklog.md after each completed task. Include: ticket ID, what was done, what was verified, any deviations from the plan.
- 8. TEST BEFORE CLAIMING DONE.** "It compiles" is never sufficient. Browser-verified interactivity is the standard. Use agent-browser to verify every change renders and works.

PART I • PHASE 0 — VERIFY FIRST

14 contradictions to settle before planning a single sprint · ~4 hours · No code changes

The 10+ source documents were written over several days. Later reports corrected earlier ones without the earlier ones being withdrawn. **14 findings now contradict each other or were never verified.** Run

each check against the live build at dz-pharm.vercel.app, record the answer, and only then proceed to the register.

V1

PWA manifest path

Contradiction: Reports #1/#8 say /manifest.json 404s. Report #5 says v8 SW live, manifest at /manifest.webmanifest.

How to settle: Fetch both paths. DevTools → Application → Manifest. Check if is in .

This plan bets: #5 is right about the file; missing tag is the real gap → P2-20

V2

Interactions Matrice render

Contradiction: Report #22: tab empty (render bug). Report #23: tested 3×3 matrix, zero bugs, grade A.

How to settle: Add DOLIPRANE + BRUFEN + LOPRIL → click Matrice. Then click Matrice with <2 drugs.

This plan bets: Both right: empty below 2-drug threshold without empty-state message → P2-11

V3

Sharing dialog

Contradiction: Report #49: no copyable URL. Report #18: full share dialog, 7/7. Master index: WhatsApp/Telegram/email/copy.

How to settle: Open medication → click Partager → screenshot. Mobile: typeof navigator.share.

This plan bets: Dialog exists; missing link inside (no URL until routing) → P2-21

V4

Toast system

Contradiction: Main report + #11/#22/#36: no toast system. UX Feedback report: working Sonner toaster, 3 types verified.

How to settle: Favourite a medication. If toast appears, toaster exists.

This plan bets: Sonner is there. Delete 'build toast system'. What remains: wire to every action + ARIA + undo → P1-05

V5

Icon button labels

Contradiction: Report #62: 4 of 6 icon-only buttons missing/generic aria-label. Report #56: all six labelled.

How to settle: Run axe DevTools. Inspect 4 buttons: theme, Botanique, Nuancier, Copilot.

This plan bets: #62 is more specific. Trust it. 25 minutes either way → P1-02

V6

Recently-viewed cap

Contradiction: Report #2: capped at 1. Report #44: 10-item cap, correct dedup.

How to settle: View 5 different medications. Read localStorage['dzpharm-store'].

This plan bets: #2 probably hit dedup-order bug (cap before dedup) → P2-09

V7

Auth bypass

Contradiction: Report #8: client-side-only, 3-second devtools bypass. Report #7: POST /api/auth is server-side.

How to settle: In fresh tab: sessionStorage.dzpharm_session = 'active' → reload.

This plan bets: #7 right on validation. But sessionStorage-as-session means client decides. Bypass shapes P0-02

V8

Lock button

Contradiction: #8: doesn't work. #9: it does — earlier test ran inside modal.

How to settle: Click from dashboard. Then from inside Copilot modal.

This plan bets: #9 right — but 'modal overlays eat header clicks' is itself a bug → P1-22

V9

Performance numbers

Contradiction: Main: TTFB 80ms, FCP 204ms, 88.8KB. Report #3: TTFB 6ms, FCP 836ms cold, 695KB.

How to settle: Lighthouse in real Chrome, mobile preset, 4G throttling, cold cache.

This plan bets: Both are edge-warm EU measurements. Re-measure before claiming 'top decile'

V10

Copilot latency

Contradiction: Quoted as 4-6s, 3-8s, and 11s in three reports.

How to settle: Time 10 queries in each of 3 modes. Record p50 and p95.

This plan bets: All three probably true at different times. Need distribution, not anecdote → P1-06

V11

Model version

Contradiction: Cmd-K says Gemini 3.8 Flash. Copilot header says 3.6.

How to settle: Read model string in the API route handler.

This plan bets: 5-minute fix, but on a clinical tool a pharmacist is entitled to know → P1-15

V12

Features #20/#21 unaudited

Contradiction: Review menu has 68 items; nothing covers #20/#21.

How to settle: Open review menu, read what #20/#21 are, reconcile against shipped product.

This plan bets: Menu is behind the product. Unbuilt or shipped-and-unaudited → P2-41

V13

Dictation safety + palette desync

Contradiction: V1 audit plan promoted these to P0 (clinical safety). Original reports had them as P1.

How to settle: Test: dictate an Arabic dose → check if transcript is correct. Test: switch palette → check if severity colours change.

This plan bets: Both are P0 — clinical safety. Dictation can produce confident nonsense doses. Palette switch can repaint clinical verdicts.

V14

Search scope (NEW in v2)

Contradiction: UX Review documents 5-scope filter (Tous/DCI/Marque/Laboratoire/N° AMM). Ultimate Blueprint P3-15 says search matches brand name only.

How to settle: Search 'paracetamol' (a DCI). If results appear, search matches DCI. If empty, only brand name.

This plan bets: Settle before building synonym layer — building on wrong assumption wastes effort.

PART II • PHASE 0.5 — DAY 1 QUICK WINS

3 independent fixes · 1 hour total · No dependencies · Ship before any pillar work

QW-01

Load Inter + Noto Sans Arabic + JetBrains Mono via next/font/google

Fix: 1. In app/layout.tsx, add: `import { Inter, Noto_Sans_Arabic, JetBrains_Mono } from 'next/font/google'`

2. Configure each: `subsets, weights, display:'swap', variable:'--font-X'`

3. Apply to

4. In globals.css: `--font-sans: '__Inter', '__Inter_Fallback', var(--font-noto-arabic), -apple-system, sans-serif`

5. Verify: `document.fonts.size === 3` in browser console

Acceptance test: `document.fonts.size === 3`. Arabic text renders in Noto Sans Arabic.

`getComputedStyle(body).fontFamily` contains `'__Inter'`.

30 MIN

DEPENDS: NONE

QW-02

Add prefers-reduced-motion CSS media query

Fix: Add to globals.css: `@media (prefers-reduced-motion: reduce) { *,*::before,*::after { animation-duration: 0.01ms !important; animation-iteration-count: 1 !important; transition-duration: 0.01ms !important; scroll-behavior: auto !important; } }`

Acceptance test: Enable 'Reduce motion' in macOS System Settings → Accessibility → Display. All animations (modal, toast, nav hover) are instant. No visible motion.

5 MIN

DEPENDS: NONE

QW-03

Add aria-labels to 4 icon-only buttons

Fix: 1. Botanique button: `aria-label='Mode botanique'`

2. Nuancier button: `aria-label='Nuancier — changer de palette'`

3. Copilot button: `aria-label='Ouvrir le Copilote IA'`

4. Theme toggle: `state-aware aria-label = {theme === 'dark' ? 'Passer en mode clair' : 'Passer en mode sombre'} + aria-pressed={theme === 'dark'}`

Acceptance test: Screen reader announces each button with its name. NVDA/VoiceOver verification. WCAG 4.1.2 passes.

25 MIN

DEPENDS: NONE (BUT VERIFY V5 FIRST)

PART III · THE REGISTER — P0 CRITICAL (8 items)

Blocks production/clinical use · Ship this fortnight · No feature work starts ahead of these ·
CLINICAL SAFETY items marked 

P0-01

No real URL routing — 22 routes return 404

Fix instructions: 1. Create app/repertoire/page.tsx with SSR + URL search params (q, status, domain, page)
2. Create app/prix-chifa/page.tsx, app/interactions/page.tsx, app/copilote/page.tsx
3. Update header: replace setActiveView() with from next/link. Active state via usePathname()
4. Create app/medicament/[slug]/page.tsx with generateStaticParams + ISR (revalidate: 86400)
5. Create intercepting route: app/repertoire/@modal/(.)medicament/[slug]/page.tsx for modal version
6. Create app/sitemap.ts listing all 9,570 URLs (static + medication)
7. Create app/medicament/[slug]/error.tsx (branded 404)
8. Create 4 supporting pages: a-propos, contact, legal, sitemap
9. Update Cmd-K palette: router.push instead of setActiveView
10. Fix Bug #1: wire single-key hotkeys via useEffect + keydown handler
11. Fix Bug #2: track selectedIndex in state, pass to Enter handler

Acceptance test: Direct URL /repertoire loads (no 404). Browser back/forward works. /medicament/doliprane-500mg-comprime loads. Sitemap lists 9,570 URLs. Cmd-K hotkeys activate on single keypress. Enter activates highlighted command (not first).

2 WEEKS

DEPENDS: NONE (BUT VERIFY V9 FIRST FOR PERFORMANCE BASELINE)

P0-02

Single-factor shared-code auth — one code for everyone, no audit trail

Fix instructions: Phase 1 (1 day): Install NextAuth.js + @auth/prisma-adapter. Create route handler with Credentials provider. Move shared code to process.env.DZPHARM_ACCESS_CODE. Replace sessionStorage with useSession(). Verify: devtools bypass closed.

Phase 2 (1 week): Prisma + PostgreSQL (Neon). User model. Registration page. Email verification. Role assignment (PHARMACIST/DOCTOR/PATIENT/ADMIN/SUPER_ADMIN).

Phase 3 (3 days): TOTP MFA via otplib (QR enrollment, backup codes, recovery).

Phase 4 (5 days): AuditLog Prisma model. logAction() helper. Instrument all clinical actions. Admin audit dashboard. 7-year retention cron.

Phase 5 (3 days): RBAC middleware (route protection). Idle timeout (15 min). Rate limiting (Upstash Redis, 5 attempts → 429).

Acceptance test: document.cookie does not contain session token (httpOnly). Users can register with email+password. MFA required for pharmacists. Every clinical action logged to userId. Audit dashboard shows recent actions. RBAC blocks unauthorized routes. Rate limiting returns 429 after 5 wrong attempts.

4 WEEKS

DEPENDS: V7 (VERIFY BYPASS EXISTS)

P0-03

Medication detail opens as modal without URL change

Fix instructions: Create intercepting route pattern: `app/repertoire/@modal/(.)medicament/[slug]/page.tsx` (modal version) + `app/medicament/[slug]/page.tsx` (full page version). Both use same data-fetching logic (`getMedicationBySlug`). Clicking a medication in `/repertoire` opens modal AND updates URL to `/medicament/[slug]`. Refresh on modal URL loads full page version.

Acceptance test: URL changes when modal opens. URL is shareable. Refresh on `/medicament/doliprane-500mg-comprime` loads full page (not modal). Both versions show same data.

4 HOURS

DEPENDS: P0-01 (ROUTING PHASE 1 MUST SHIP FIRST)

P0-04

🚑 CLINICAL SAFETY

🚑 Voice dictation hard-wired to fr-FR — Arabic doses become confident nonsense

Fix instructions: 1. Detect `!window.SpeechRecognition` on mount → render disabled button with tooltip 'Dictée non supportée sur Firefox'

2. Add [FR/AR] manual language toggle next to mic button → sets `recognition.lang`

3. Add permission pre-prompt modal on first click: 'DzPharm utilise votre microphone pour transcrire votre message. L'audio n'est jamais envoyé ni stocké.'

4. Fix stop button aria-label: conditional `isRecording ? 'Arrêter la dictée' : 'Dicter un message'`

5. Add mandatory review toast after dictation: 'Relisez votre message avant d'envoyer' + 500ms delay before send button enables

6. Defer Darija (Whisper server-side) to Phase 3

Acceptance test: Firefox: button shows 'non supporté' tooltip. Arabic toggle switches `recognition.lang` to 'ar-DZ'. Permission pre-prompt appears on first click. Stop button aria-label changes with state. Review toast appears after dictation. 500ms delay before send.

6 HOURS

DEPENDS: V13 (VERIFY CURRENT DICTATION BEHAVIOR)

P0-05

🚑 CLINICAL SAFETY

🚑 Copilot does arithmetic — LLM computes doses with no deterministic cross-check

Fix instructions: 1. Create a deterministic dose-verification engine (`lib/dose-engine.ts`) that independently computes: $\text{mg/kg} \times \text{weight} = \text{dose in mg}$ → dose in mL/sachets

2. After Copilot response, run the engine on the same input (drug, weight, age)

3. If engine result differs from Copilot output by >5%: display warning '⚠️ Vérification: le calcul indépendant donne X mg. Le Copilote a indiqué Y mg. Vérifiez manuellement.'

4. If engine result matches: display small '✓ Vérifié' badge

5. The engine handles: paracetamol, ibuprofen, amoxicillin, and all drugs in the posology calculator's 15-molecule list

6. Never let the LLM be the sole source of a dose calculation

Acceptance test: Copilot response includes '✓ Vérifié' badge when dose matches engine. Warning displayed when dose differs. Engine independently computes correct dose. No dose is presented without verification.

3 DAYS

DEPENDS: NONE (CAN START IMMEDIATELY)

P0-06

CLINICAL SAFETY

Cmd-K selection desync — Enter can dial the wrong emergency service

Fix instructions: 1. Track selectedIndex in useState (default 0)

2. ArrowDown: setSelectedIndex(i ⇒ Math.min(i+1, filteredOptions.length-1))

3. ArrowUp: setSelectedIndex(i ⇒ Math.max(i-1, 0))

4. Enter: const selected = filteredOptions[selectedIndex]; if selected.action → selected.action(); if selected.href → router.push(selected.href); if selected.tel → window.location.href = `tel:\${selected.tel}`

5. Ensure visual highlight matches selectedIndex (aria-selected on the correct option)

Acceptance test: ArrowDown highlights second item. Enter activates highlighted item (not first). SAMU 14 is only called when SAMU is highlighted. Anti-Poison is called when Anti-Poison is highlighted.

3 HOURS

DEPENDS: NONE

P0-07

CLINICAL SAFETY

Nuancier palettes may repaint clinical severity colours

Fix instructions: 1. Add CSS rule: .severity-safe, .severity-warning, .severity-danger, .severity-severe, .severity-info, .severity-none, .severity-unknown { color: var(--state-safe) !important; /* etc */ } — these classes are immune to palette overrides

2. Audit all 20 Nuancier palettes: verify none override --state-* variables

3. In the switchPalette() function: only override --primary, --accent, --secondary, --muted. Never --state-*

4. Add a test: switch to each palette, verify severity colours unchanged

Acceptance test: Switching to any of 20 palettes does not change the 7 severity colours. Severity badges remain green/amber/red/orange/cyan/slate/purple regardless of active palette.

4 HOURS

DEPENDS: NONE

P0-08

Login form error announcement needs verification

Fix instructions: 1. Run V7 verification first

2. If login HAS role=alert + aria-live=assertive (as #7 report found): mark P0-04 as RESOLVED, focus on the 4 missing live regions (P1-08)

3. If login does NOT have role=alert: add

region above the form. On error, set text content. Set aria-invalid='true' on input. Set aria-describedby='pw-error' linking input to error.

Acceptance test: Screen reader announces 'Mot de passe incorrect' immediately on wrong code. aria-invalid=true on input. aria-describedby links input to error. WCAG 3.3.1 passes.

30 MIN (IF MISSING) / 0 (IF ALREADY PRESENT)

DEPENDS: V7 (VERIFY CURRENT STATE)

PART III · THE REGISTER — P1 HIGH PRIORITY (28 items)

Materially harms UX, trust, or brand · Next sprint cycle (weeks 1-6) · Ship in parallel with P0

The 28 P1 items cover: typography (webfont loading), accessibility (icon labels, live regions, RTL), Copilot bugs (streaming, history, mode patient, version, prompts), state management (drug list persistence, cross-tab sync), session (idle timeout, lock behavior), and tool fixes (Chifa routing bug,

shortage submit 500, interactions list). Each has a fix instruction and acceptance test in the full register.
The 10 highest-impact P1 items:

P1-01

Load Inter + Noto Sans Arabic via next/font

Same as QW-01 (if not shipped as quick win)

30 MIN

P1-02

Add aria-labels to 4 icon-only buttons

Same as QW-03 (verify V5 first)

25 MIN

P1-05

Wire Sonner toasts to every mutating action + ARIA + undo

Verify V4 first. Toast exists; needs wiring.

2 HOURS

P1-06

Adopt Vercel AI SDK for streaming Copilot responses

streamText + useChat. 10× perceived latency improvement.

1 DAY

P1-08

Add 4 aria-live regions (Copilot, toast, bell, search)

role=status announcing async events to screen readers

2 HOURS

P1-09

Wire Copilot suggested-prompt onClick handlers

setInput(prompt.text) + sendMessage(prompt.text)

1 HOUR

P1-10

Add Copilot conversation history to Zustand (persist)

copilotMessages slice. Survives refresh. Prune: last 30.

1 DAY

P1-11

Fix Mode patient = Mode pro (upgrade system prompt)

Add MODE_PROMPTS.patient section to buildSystemPrompt()

4 HOURS

P1-12

Persist interactions drug list in URL params

useInteractionListState hook. ?drugs=8421,1523. Survives refresh.

4 HOURS

P1-14

Add dir=rtl + lang=ar to all Arabic content elements

Arabic text renders right-to-left. Question mark on correct side.

25 MIN

P1-15

Fix Copilot version label inconsistency (3.8 vs 3.6)

Verify V11. Update both locations to match deployed model.

5 MIN

P1-16

Fix Chifa simulator: adding drug routes to Comparator (shared handler bug)

Verify the onClick handler calls Chifa add, not Comparator add

3 HOURS

P1-22

Fix: modal overlays eat header button clicks (lock button)

z-index audit. Ensure header buttons are above modal backdrop.

2 HOURS

P1-24

Fix shortage report submit returns HTTP 500

Check /api/shortage endpoint. Likely missing CSRF or DB connection.

4 HOURS

P1-26

Buy domain (dzpharm.dz or chosen name) BEFORE submitting sitemap

Buy in Week 1. Never submit 9,570 URLs from *.vercel.app and re-submit after domain move.

1 DAY

Remaining 13 P1 items (P1-03 input border, P1-04 H1 tagline, P1-07 chunked TTS, P1-13 reduced motion [may be QW-02], P1-17 parapharmacy filter returns 0 rows, P1-18 100/page selector renders 40, P1-19

«Copier le plan» fails in Ramadan, P1-20 search result count missing, P1-21 breadcrumb on nested views, P1-23 enterkeyhint on login, P1-25 Copilot loading indicator, P1-27 dashboard max-width, P1-28 family cabinet audit gap) — each with fix instructions in the full register (DzPharm_Master_Audit_Plan_v2.html, Section 04).

PART III · THE REGISTER — P2 MEDIUM (42 items) & P3 LOW (44 items)

P2: degrades polish, reliability, governance · This quarter · 42 items
P3: enhancement-grade · Backlog, each <4h · 44 items

The 42 P2 items and 44 P3 items are documented in full in the uploaded Master Audit Plan v2 (Section 04, the complete 122-item register). Key P2 items include: palette contrast verification (P2-18), Storybook setup (P2-19), manifest link tag (P2-20), recently-viewed cap fix (P2-09), cross-tab sync (P2-10), Matrice empty-state (P2-11), idle timeout (P2-15), density mode switch (P2-39), stat drift fix — generate counts from dataset not hardcoded (P2-28), measurement stack — analytics + error tracking + uptime (P2-25/26). Key P3 items include: wilaya-specific emergency numbers, print CSS, tri-state theme, hreflang, og:image, fuzzy search, DCI search, pharmacy GPS coordinates, text-only palette, editorial serif, bottom nav for mobile, swipe gestures.

FULL REGISTER REFERENCE

The complete 122-item register with fix instructions, acceptance tests, effort estimates, and dependencies for ALL items (P0 through P3) is in the uploaded file: **DzPharm_Master_Audit_Plan_v2.html** (Section 04). This document provides the execution sequence; that file provides the per-item detail. An AI agent executing this plan should read both documents: this one for the *order* and *rules*, that one for the *instructions*.

PART IV · FEATURE VERDICTS — 24 Original Ideas Assessed

Each of the 24 ideas from the UX Review assessed: ship / defer / kill / merge

#	IDEA	VERDICT	NOTES
F1	Darija voice-first Copilot	SHIP (LONG BET)	Whisper STT + Gemini + Darija TTS. After P0-04 dictation fix + P0-05 dose engine. 8 weeks.
F2	Prescription OCR scanner	SHIP (NEXT)	Camera → OCR → Copilot cross-check. After P0-01 routing. 6 weeks.
F3	Drug-image recognition	DEFER	Requires medication image database (#N29). 8 weeks after database.

F4	Personalized AI formulary per pharmacy	SHIP NOW	Pharmacy uploads stock CSV. Copilot suggests substitutes. 3 weeks. Force-multiplier.
F5	AI-powered prescription translation FR → AR	SHIP NOW	Auto-translate for patient counseling. 2 weeks. Critical for 60% of patients.
F6	Multi-turn patient case solver	SHIP (NEXT)	After P1-10 (Copilot history) + P0-02 (auth). 3 weeks.
F7	Renal-dose adjustment calculator	SHIP (NEXT)	Expand existing Cockcroft-Gault. 3 weeks.
F8	Hepatic dose adjustment calculator	SHIP (NEXT)	Child-Pugh → dose. 3 weeks. Completes organ-impairment pair.
F9	Antibiogram by region	DEFER	Requires crowdsourced data pipeline. 8 weeks after shortage center fixed.
F10	Pregnancy decision tree (trimester-by-trimester)	SHIP (NEXT)	Expand existing #10 checker. 2 weeks.
F11	Pediatric weight-band dosing tables	SHIP NOW	WHO-style cards, printable A5. 2 weeks. DO NOW.
F12	IV compatibility matrix	SHIP (LONG BET)	Y-site checker. 6 weeks. No competitor in Algeria.
F13	Real-time shortage heatmap	SHIP (NEXT)	After P1-24 (fix shortage submit). 4 weeks. Builds on #50.
F14	Price-history chart per medication	FILL IN	PPA history over 24 months. 2 weeks. Needs data collection starting now.
F15	Pharmacy inventory marketplace	DEFER	Requires #8 auth + trust infrastructure. 8 weeks.
F16	National pharmaceutical statistics dashboard	DEFER	Requires data governance. 8 weeks after auth.
F17	Per-user accounts with role-based access	ALREADY PLANNED	This IS P0-02. The foundation. 4 weeks.
F18	Saved medication lists & collections	SHIP (NEXT)	After #8 auth. Builds on #48 favorites. 3 weeks.
F19	Cross-device sync via account	SHIP (NEXT)	After #8 auth + server-side state. 2 weeks.
F20	CME credit tracking	FILL IN	After #8 auth + audit log. 2 weeks. Subject to medical council.
F21	Native iOS/Android apps	SHIP (LONG BET)	Capacitor wrap of PWA. 8 weeks after P0-01 routing.
F22	WhatsApp bot integration	SHIP NOW	Bot answers quick queries. 3 weeks. DO NOW. Force-multiplier.

F23	Public REST API for researchers	DEFER	After data governance. 6 weeks.
F24	Open data exports (CC-BY-SA)	DEFER	After data governance. 4 weeks.

PART V · THE INNOVATION CATALOG — 200+ Ideas

From 5 source documents, deduplicated and classified · Tagged NOW / NEXT / LONG BET / FILL-IN
· 93 marked ★ NEW in v2

The complete innovation catalog (200+ ideas across 16 domains) is in the uploaded file: **DzPharm_Innovation_Vision_Brainstorm_v2.html**. This document provides the execution sequence; that file provides the per-idea detail. The 6 «Big Swings» that define the innovation vision:

BS-01

The Compatibility Layer

Halal/haram, alcohol-content, and Ramadan-fasting compatibility flag on every one of the 9,555 records. The one clinical dimension no European/American/Gulf platform will build correctly for Algeria. Not a translation problem — a jurisprudence problem.

NEXT · L

GATE: NONE (DATA WORK CAN START NOW)

BS-02

A Copilot That Speaks First

Today the AI only answers. It could notice: expiring EpiPen in Family Cabinet, watched shortage back in stock, Ramadan schedule needs re-timing — and say so before being asked. A proactive «Pour vous» panel.

NEXT · M

GATE: P1-10 (HISTORY), RECALL FLAGS

BS-03

The Counter, Not the Tab

Barcode scan → Chifa card read → interaction check → printed handout, as one motion at the point of dispensing. DzPharm becomes the software the counter runs on.

LONG BET · XL

GATE: P0-01 ROUTING, P1-16 CHIFA REPAIR

BS-04

★ NEW IN V2

★ NEW: The Official-Price Guardian

Patient-facing page — no login — where anyone types a medication and sees the official PPA price, Chifa rate, and answers «est-ce normal?». The mass-market trust channel. Virality built in: screenshots travel through family WhatsApp groups.

NEXT · M

GATE: P0-01 (PUBLIC ROUTE), P2-27 (FRESHNESS)

BS-05 ★ NEW IN V2

★ NEW: The Safety Net — Dispensing-Error Elimination

LASA (look-alike/sound-alike) pairs with Tall-Man lettering. High-alert double-check (insulin, methotrexate, anticoagulants) with dose restated in words. Duplicate therapy auto-caught. P0-08 dose engine is the backbone.

START NOW · M PER LAYER

GATE: P0-08 ENGINE

BS-06 ★ NEW IN V2

★ NEW: Self-Healing Data — The Moat That Maintains Itself

ANPP/Ministry regulatory-sync bot. Visible freshness (lastVerifiedAt badges). Correction-request pipeline where pharmacists fix records and earn credit. Weekly price snapshots. Clinical eval harness (200 expert answers, ≥95% forever).

NEXT · L STAGED

GATE: P2-27 SKELETON FIRST, P0-02 FOR CREDITED CORRECTIONS

The remaining 194 ideas span 16 domains: Design Language (50+ palette directions, 25+ custom symbols, motion, sound, print), Intelligence Layer (AI interaction severity, dosage individualization, adherence coach, natural language search, citations, multi-language prescriptions, duplicate therapy detection), Clinical Tools (warfarin INR, anticoagulation reversal, controlled substance tracking, allergy cross-reactivity, pharmacogenomics, medication reconciliation), Search Reimagined (fuzzy, DCI-based, synonym, voice, visual graph), Community & Supply (shortage heatmap, inventory marketplace, pharmacy directory), Data & Business (equivalence graph, real-time CNAS, medication images, patient profile, shortage prediction), Inclusive Design (Tamazight, low-literacy, sign language, dyslexia font, high contrast), Beyond Browser (WhatsApp bot, native apps, Alexa/Google Home, AR), and 15 Moonshots (AR medication identification, blockchain authenticity, telepharmacy, federated learning, gamified training). **All 200+ ideas with full descriptions, impact assessments, and effort estimates are in [DzPharm_Innovation_Vision_Brainstorm_v2.html](#).**

PART VI · THE DESIGN CONSTITUTION

The rules that govern every design decision · From Design Universe + Design Exploration + Master Audit Plan

The complete design constitution (live interactive version with palette swatches, symbol designs, box variants, border weights, motion curves) is in the uploaded file: **[dzpharm-design-exploration.html](#)**. The key rules that an AI agent MUST follow:

DC-01

Severity colours are LOCKED

The 7 clinical severity colours (safe/warning/danger/severe/info/none/unknown) are immune to palette overrides. No palette, theme, or animation may change them. This is P0-07. Enforcement: CSS !important on severity classes, switchPalette() never touches --state-* variables.

DC-02

Lightness-step palette method

Rather than picking 9 colours by eye, step one hue through consistent lightness increments. Dark mode: canvas L8-10%, surface L14-16%, raised L20-22%, ink L92%+. Light mode: invert. Contrast is correct by construction.

DC-03

9 palette roles

Every palette defines: Canvas, Surface, Surface-raised, Ink-primary, Ink-secondary, Ink-tertiary, Border, Accent, Accent-soft. The roles never change; only the hue poured into them changes.

DC-04

70/25/5 accent rationing

In any view: 70% neutral (canvas/surface/ink), 25% brand (primary/accent), 5% semantic (state colours). If more than 5% of pixels are severity colours, the design is shouting — reduce.

DC-05

Motion communicates or is cut

Every animation must answer a question («did my click work?»). 150-250ms sweet spot. Fast in, slow out: cubic-bezier(0.16,1,0.3,1). prefers-reduced-motion respected absolutely. The beacon-ping (emergency dot pulse) is reserved for genuine urgency forever.

DC-06

One signature loader

Replace every generic spinner with a single two-tone capsule motif that splits and rejoins. Used everywhere instead of reinvented per screen. Skeletons (shape of arriving content) for tables/lists; capsule only for click-triggered waits.

DC-07

12 box variants

Standard, Accent, Glow, Glassmorphism, Neumorphism, Outline, Danger, Success, Warning, Info, Minimal, Hero. Each with CSS spec. Card architecture: header/body/footer. 4px radius scale (2/6/12/16/24/32px).

DC-08

Whitespace = 4px base

3 density modes: Compact (8-12px padding, 6px gap), Standard (16-20px, 12px), Spacious (24-32px, 20px). Max-width: 80rem on main content. The luxury of space: premium products use MORE whitespace, not less.

DC-09

15 custom DzPharm symbols

chifa-card, pharmacy-cross-algeria, mortar-pestle, ramadan-crescent, list-one, list-two, amine-order, medication-bottle-dz, darija-translate, wilaya-pin, barcode-scan-dz, interaction-cross, posology-scale, pregnancy-shield, shortage-alert. All 2px stroke, 24×24 grid, matching Lucide.

DC-10

Typography = Inter + Noto Sans Arabic + JetBrains Mono

8-step modular type scale with `calc()` line-heights. Variable fonts (1 file per family). `font-display: swap`. `document.fonts.size === 3` after loading. This is QW-01.

PART VII · THE EXECUTION ROADMAP — 8 Phases

13 weeks · 2 engineers + 1 designer + 1 PM · Never reorder · Dependencies are hard gates

Phase 0

Verify First · 4 hours

Work: 14 contradiction checks (V1-V14). No code changes. Record answers. This is the highest-value half day in the plan.

Done when: All 14 checks resolved. Answers recorded. Contradictions settled.

Phase 0.5

Day 1 Quick Wins · 1 hour

Work: QW-01 (next/font), QW-02 (reduced-motion CSS), QW-03 (aria-labels). Ship in one PR before any pillar work.

Done when: `document.fonts.size === 3`. WCAG AAA motion. WCAG 4.1.2 icon labels.

Phase 1

P0 Safety Cluster · Weeks 1-4

Work: P0-01 routing, P0-02 auth, P0-03 modal URL, P0-04 dictation, P0-05 dose engine, P0-06 Cmd-K desync, P0-07 severity lock, P0-08 login a11y. NO feature work starts ahead of these. In parallel: P1-01 (webfont), P1-02 (icon labels), P1-06 (streaming), P1-09 (prompts), P1-11 (mode patient), P1-12 (drug list URL), P1-14 (RTL), P1-15 (version label).

Done when: All 8 P0 items resolved. Devtools bypass closed. 9,570 routes work. Per-user auth with MFA. Dose engine verifies Copilot. Severity colours locked. Cmd-K selection correct.

Phase 2

P1 Polish + Tools · Weeks 5-8

Work: Remaining P1 items: P1-05 (toast wiring), P1-08 (live regions), P1-10 (Copilot history), P1-16 (Chifa fix), P1-22 (modal z-index), P1-24 (shortage 500 fix), P1-26 (domain purchase). Start P2 items in parallel: P2-09 (recently-viewed cap), P2-10 (cross-tab sync), P2-18 (palette contrast), P2-19 (Storybook), P2-25/26 (analytics + error tracking).

Done when: All 28 P1 items resolved. Toasts wired to every action. Live regions announce async events. Copilot history persists. Chifa works. Shortage submit works. Domain purchased. Analytics installed.

Phase 3

P2 Medium + Feature NOW · Weeks 9-10

Work: Remaining P2 items (42 total, ~20 remaining). Start NOW-tagged features: F4 (formulary upload), F5 (prescription translation), F11 (weight-band cards), F22 (WhatsApp bot), BS-05 (LASA safety net, start now).

Done when: All P2 items resolved. 5 NOW features shipped. LASA look-alike/sound-alike warnings live.

Phase 4

P3 Backlog + Feature NEXT · Weeks 11-12

Work: P3 items (44, each <4h, sprint fillers). NEXT features: F2 (OCR scanner), F6 (multi-turn case solver), F7 (renal calculator), F8 (hepatic calculator), F10 (pregnancy decision tree), F12 (IV matrix), F13 (shortage heatmap), F18 (saved lists), F19 (cross-device sync). Big Swings: BS-01 (compatibility layer), BS-02 (proactive Copilot), BS-04 (price guardian), BS-06 (self-healing data).

Done when: P3 backlog cleared. 10 NEXT features shipped. 4 big swings initiated.

Phase 5

Design System A+ · Week 12 (parallel)

Work: Storybook complete (all 400 tokens, 15 symbols, 12 box variants, 9 shadows, 50+ palettes). CI checks (axe-core contrast, color-blindness). Per-palette per-theme verification. Text-only palette. Governance docs. This runs in parallel with Phase 3-4.

Done when: Storybook deployed. CI prevents contrast regression. All 50+ palettes verified. Design system grade: A+.

Phase 6

Integration + Rollout · Week 13

Work: E2E testing (user journey, SEO crawl, a11y, OWASP). Feature flag rollout: 10% → 50% → 100%. Remove deprecated code. Submit sitemap to Google. Monitor 1 week.

Done when: 100% rollout. Error rate <0.05%. Sitemap submitted. Platform grade: A+. Zero D-grade features. Zero P0 issues.

PART VIII · DEFINITION OF DONE + RISK REGISTER + KPIs

Platform-level completion criteria · 8 risks · 10 KPIs to measure

8.1 Definition of Done — Platform Level

1. All 9,570 routes return 200 (no 404s on valid routes). Sitemap submitted to Google.

2. Users register with email + password + TOTP MFA. Shared code deprecated. Devtools bypass closed.

3. Every clinical action logged to userId with 7-year retention. Audit dashboard live. Law 85-05 compliant.

4. Drug list survives refresh (URL params). Copilot history persists (Zustand + server). Favorites sync across tabs and devices.

5. Copilot streams responses (~1s first token). Mode patient produces simplified output. Darija queries get Darija responses.

6. Dose engine independently verifies every Copilot dose calculation. '✓ Vérifié' badge on matching doses. Warning on mismatch.

7. Severity colours locked across all 50+ palettes. No palette can repaint clinical verdict colours.

8. Inter + Noto Sans Arabic loaded (document.fonts.size === 3). Brand identity stable across all devices.

9. Storybook documents all tokens, components, states. CI prevents contrast/color-blindness regression.

10. prefers-reduced-motion respected. All icon-only buttons have aria-labels. 4 aria-live regions announce async events. WCAG 2.2 AA passes.

11. Feature flags removed. Error rate <0.05%. Analytics + error tracking live. Status page operational.

12. Overall platform grade: A+. Zero D-grade features. Zero P0 issues. Zero P1 issues.

8.2 Risk Register (Top 8)

#	RISK	PROBABILITY	IMPACT	MITIGATION
1	Routing migration breaks setState navigation	Medium	High	Feature flag, keep old code as fallback, 10% → 50% → 100% rollout
2	NextAuth migration breaks existing sessions	Medium	High	Phase 1 keeps shared code (transitional), gradual migration
3	Gemini API cost exceeds budget at scale	Low	Medium	Cost optimizations: pruning, caching (15-25% hit rate), model selection, token budgeting
4	9,555 pages cause Vercel build timeout	Medium	Medium	Use ISR (revalidate: 86400), not SSG. Pages generate on first request, not at build.
5	Noto Sans Arabic increases payload beyond acceptable	Low	Low	Variable font (1 file), SW cache, font-display: swap. LCP stays under 300ms.
6	License verification API (Order of Pharmacists) unavailable	High	Medium	Admin manual review fallback. Auto-approve if API exists, manual if not.
7	Team capacity (2 engineers, 13 weeks)	Medium	High	Day 1 quick wins ship regardless. Parallel tracks. Prioritize safety cluster.
8	Domain not purchased before sitemap submission	Medium	High	Buy domain in Week 1 (P1-26). Never submit 9,570 URLs from *.vercel.app and re-submit after move.

8.3 KPIs to Measure

KPI	CURRENT	TARGET	HOW TO MEASURE
Indexable pages	1	9,570	Google Search Console → Coverage
Monthly organic visits	~500	~190,000	Google Analytics 4
Copilot perceived latency (p50)	3-11s	~1s	Client-side timing: submit → first token
Platform grade	B-	A+	Re-run 68-feature audit
Error rate	Unknown	<0.05%	Sentry / Vercel error tracking
Copilot cost per query	~\$0.0008	~\$0.0004	Gemini API usage dashboard
User registrations	0	Growing	Prisma user count
WCAG 2.2 compliance	Partial (fails 4.1.2, 4.1.3)	AA pass	axe-core CI + manual NVDA test
Audit trail coverage	0%	100% of clinical actions	Prisma AuditLog count vs action count
document.fonts.size	0	3	Browser console

THE FINAL INSTRUCTION

This document is the execution sequence. The uploaded files are the per-item detail:

- **DzPharm_Master_Audit_Plan_v2.html** — the 122-item register with fix instructions, acceptance tests, dependencies, and the 8-phase execution plan. Read Section 04 for per-item detail.
- **DzPharm_Innovation_Vision_Brainstorm_v2.html** — the 200+ idea catalog with descriptions, impact, effort, and timing tags. Read for feature implementation detail.
- **dzpharm-design-exploration.html** — the live interactive design constitution with palette swatches, symbol designs, box variants, border weights, motion curves. Read for design execution.
- **dzpharm-audit-plan.html** and **dzpharm-innovation-brainstorm.html** — prior versions, superseded by v2 but included for reference.

Execution order: Phase 0 (verify) → Phase 0.5 (quick wins) → Phase 1 (P0 safety) → Phase 2 (P1 polish) → Phase 3 (P2 + NOW features) → Phase 4 (P3 + NEXT features) → Phase 5 (design A+) → Phase 6 (rollout). Never reorder. Never skip verify-first. Never start feature work before P0 safety cluster. **The ceiling is the moon. Start with Phase 0 today.**

The Complete Document Set

This document synthesizes **~1,000+ pages** of audit material from 15 source documents into one executable plan. The complete set:

- 5 uploaded v2 HTML files (this document's primary sources)
- 59 deep-dive PDF reports (~600 pages, all 68 features)
- 6 technical design documents (169 pages, 6 pillars)
- Platform Transformation Blueprint (21 pages)
- Executive Presentation (15 slides)
- Sprint Planning Document (19 pages, 77 tickets)
- Interactive Dashboard (Next.js web app)
- Ultimate Audit & Innovation Blueprint (32 pages)
- The Design Universe (22 pages)
- Name & Homepage Vision (23 pages)
- **This document — THE FINAL EXECUTION PLAN**

Total: **~1,000+ pages of audit, design, and planning material**, now distilled into one sequenced, classified, executable plan. Ship Phase 0 today.

A+

122 register items. 200+ feature ideas. 14 verify-first checks. 24 feature verdicts. 10 design rules. 8 execution phases. 12 definition-of-done criteria. 8 risks. 10 KPIs. One plan. One sequence. One ceiling: the moon.

8

P0 CRITICAL

28

P1 HIGH

42

P2 MEDIUM

44

P3 LOW